

Two advantages of an experiment over an observational study:

- 1) We can study the specific factors we are interested in while controlling the effects of *lurking variables*.
- 2) Experiments also allow us to study the combined effects of several factors.

## How to Design an Experiment

- *Factor* – THE EXPLANATORY VARIABLES IN THE EXPERIMENT
- *Level* – THE VARIOUS GROUPS THE FACTORS TAKE

*Factor vs Level (CAREFUL)* If an experiment compared the drug dosages 50 mg, 100 mg, and 150 mg, then the factor "drug dosage" would have three levels: 50 mg, 100 mg, and 150 mg. Be sure not to confuse the number of levels of a factor with the number of factors in an experiment. Drug dosage is one factor consisting of three levels.

## Principles of Experimental Design

Keep in mind the 4 critical elements of experimental design:

### 1. COMPARISON

- We want to make sure we are using a design that compares two or more treatments. We need to make sure that the groups we are comparing though don't differ greatly before our experiment begins or bias can result. To avoid this, we use....

### 2. RANDOMIZATION

- The most important element of any experiment. It must be incorporated either in the selection process of experimental units and/or the distribution of experimental units into treatment and control groups. You can use your calculator, the random digit table or names out of a hat or flipping a coin to randomize an experiment.
  - We can only compare when the groups are as similar as possible before the treatment is applied.
  - Randomization produces groups of experimental units we expect to be similar in all respects before treatment is applied.
  - Therefore, measured differences must be due either to treatment or by chance of random assignment.

### 3. CONTROL

- CONTROL group is treated identically in all respects to the group receiving the treatment except that the members of the control group do not receive the treatment.
  - The control group is our baseline and our experimental group has only one thing changed- the explanatory variable.
  - This reduces variability in the response variable. If one group is controlled, we would expect their responses to be controlled as well.

4.

# REPLICATION

- Use enough experimental units in each group so that any difference in the effects of the treatments can be distinguished from chance differences between groups. In statistics, replication means “use enough subjects”, whereas in other fields replication means repeating.
  - Even with control, natural variability occurs among experimental units.
  - We would like to see units within a treatment group responding similarly to one another, but differently from units in other treatment groups (then we can be sure that the treatment is responsible for the differences).
  - If we assign many individuals to each treatment group, the effects of chance (and individual differences) will average out.

## Experimental Terms

- **Placebo:** A “DUMMY” TREATMENT
- **Placebo Effect:** SUBJECTS RESPOND TO THE DUMMY TREATMENT IN WAYS WE WOULD EXPECT THEM TO RESPOND TO THE ACTUAL
- **Blinding:** WHEN THE SUBJECT DOES NOT KNOW THE TREATMENT THEY’RE RECEIVING TO REMOVE THE POWER OF SUGGESTION.
- **Double Blind:** THE SUBJECT AND THE ADMINISTRATOR ARE BOTH UNAWARE OF THE TREATMENT BEING GIVEN.

## Experimental Design

The design of an experiment is crucial. Experiments are designed with the purpose of isolating the effect of the treatment on the response variable and removing any confounding effects.

- In a poorly designed experiment, IT MIGHT BE DIFFICULT TO TELL IF THE EXPLANATORY VARIABLE CAUSED A CHANGE IN THE RESPONSE OR IF IT WAS ANOTHER VARIABLE NOT MEASURED.
- Confounding variables are variables that might affect the outcome, BUT WE DID NOT CONTROL OR ACCOUNT FOR THEM IN OUR EXPERIMENT.

One way that we have seen already of removing the effect of any confounding variables is to randomly assign subjects to the treatment or control group. This way any possible bias in the population should be evenly spread among the treatment and control groups. Sometimes instead of relying on randomization to make the groups as even as possible we actually force the groups to be similar.

**Completely Randomized Design:** the experimental units are assigned to the treatments completely by chance.

- TREATMENT GROUPS AND CONTROL GROUPS WILL BE ROUGHLY EQUAL IN SIZE
- BECAUSE MATH (STAY TUNED)

Example) Does regularly taking aspirin help protect people against heart attacks? The Physicians' Health Study was a medical experiment that helped answer this question. In fact, the Physicians' Health Study looked at the effects of two drugs: aspirin and beta-carotene. Researchers wondered whether beta-carotene would help prevent some forms of cancer. The subjects in this experiment were 22,000 male physicians. There were two explanatory variables (factors), each having two levels: aspirin (yes or no) and beta-carotene (yes or no). Combinations of the levels of these factors form the four treatments in the picture. One-fourth of the subjects were assigned at random to each of these treatments.

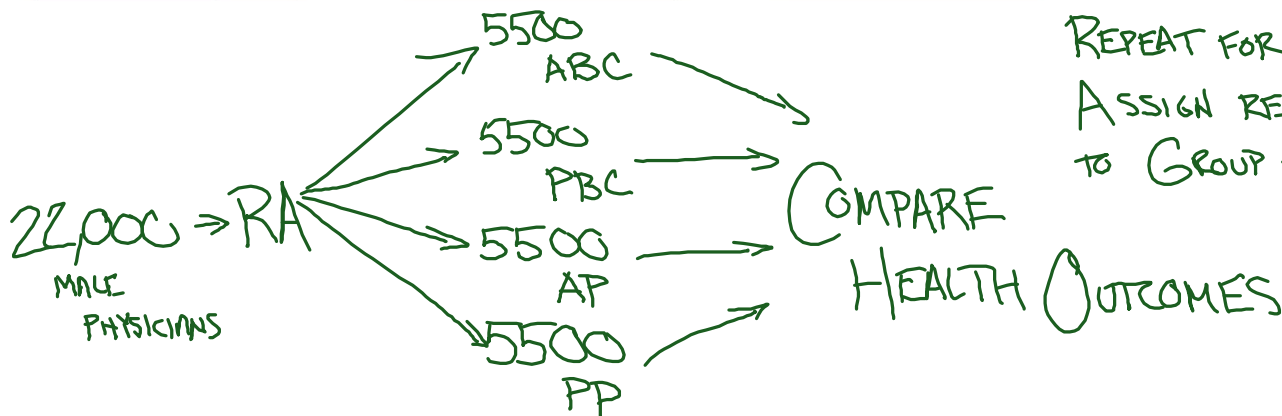
On odd-numbered days, the subjects took either a tablet that contained aspirin or a *dummy pill that looked and tasted like aspirin but has no active ingredient* (a placebo). On even-numbered days, they took either a capsule containing beta-carotene or a placebo.

There were several response variables – the study looked for heart attacks, several kinds of cancer, and other medical outcomes. After several years, 239 of the placebo group but only 139 of the aspirin group has suffered heart attacks. This difference is large enough to give good evidence that taking aspirin does reduce heart attacks. It did not appear, however, that beta-carotene had any effect on preventing cancer.

|                   |     | Factor 2: Beta-carotene                                                                        |                                                                                                      |                                                                                                |                                                                                                |
|-------------------|-----|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|                   |     | Yes                                                                                            |                                                                                                      | No                                                                                             |                                                                                                |
| Factor 1: Aspirin | Yes | Aspirin<br> | Beta-Carotene<br> | Aspirin<br> | Placebo<br> |
|                   | No  | Placebo<br> | Beta-Carotene<br> | Placebo<br> | Placebo<br> |

Describe and design a completely randomized experiment.

RA (RANDOM ASSIGNMENT): WRITE EACH OF THE 22,000 MALE NAMES ON PIECES OF PAPER, PUT THEM IN A BAG, AND MIX IT UP. THE FIRST 5500 PIECES YOU SELECT, WITHOUT REPLACING, ASSIGN TO GROUP 1. REPEAT FOR GROUPS 2 AND 3. ASSIGN REMAINING PIECES TO GROUP 4.



*AP Exam Tip: If you are asked to describe the design of an experiment on the AP exam, you won't get full credit for providing only a diagram like the one above. You are expected to describe how the treatments are assigned to the experimental units and to clearly state what will be measured or compared.*

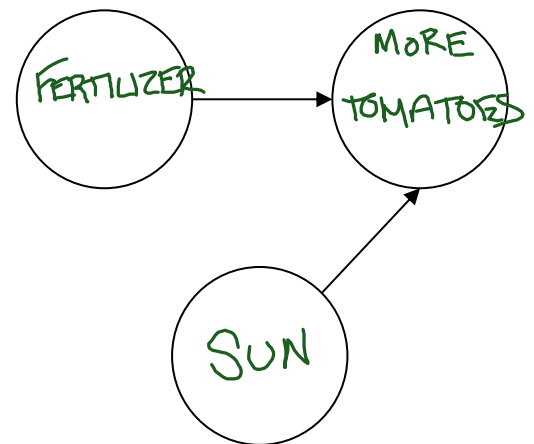
*Make sure the method is described in sufficient detail. For example, saying "assign the subjects to the groups using random digits" isn't sufficiently detailed because there are many ways to use random digits. If you use random digits, you must use labels of the same length (01-30 and not 1-30), and you must state how you address repeated numbers or numbers outside the range.*

## **Lurking vs Confounding**

An *extraneous* variable is one that is not one of the explanatory variables in the study, but is thought to affect the response variable. There are two types of extraneous variables present in studies:

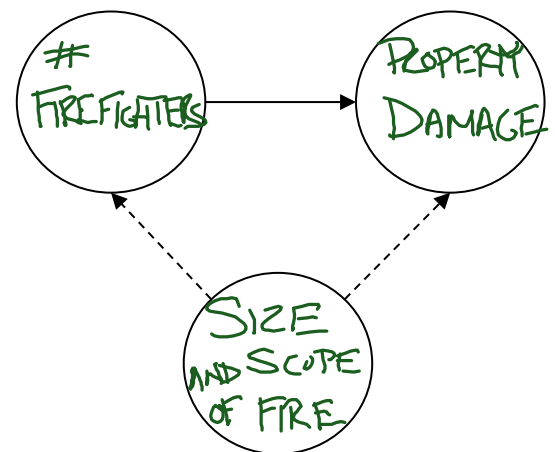
**Confounding** refers to a problem that can arise in an experiment, when there is another variable that may affect the response and is in some way tied together with the factor under investigation, leaving us unable to tell which of the two variables (or perhaps some interaction) caused the observed response.

For example, we plant tomatoes in a garden that's half-shaded. We test a fertilizer by putting it on the plants in the sun and apply none to the shaded plants. Months later the fertilized plants bear more and better tomatoes. Why? Well, maybe it's the fertilizer, maybe it's the sun, maybe we need both. We're unable to conclude that the fertilizer works because any effect of fertilizer is confounded with any effect of the extra sunshine.



**Lurking** variables are sometime referred to as "common response". That's where some other variable drives each of the two variables under investigation, making it appear that there's some association between those two variables.

For example, there is a strong association between the number of firefighters who respond to a fire and the amount of damage done. One shouldn't conclude that the firefighters may be responsible for the damage; the lurking variable is the size of the fire.



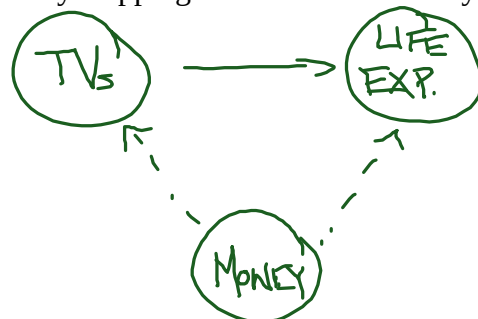
Lurking variables are the risk we face in sampling and observational studies. In an experiment, though, the factor under consideration isn't being driven by some lurking variable, because we are the ones in control there.

**Directions:** For each problem, state whether the relationship between the two variables involves causation, lurking variable, or a confounding variable.

**Problem # 1:**

Does TV make you live longer? Measure the number of television sets per person  $x$  and the average life expectancy  $y$  for the world's nations. There is a high positive correlation: nations with many TV sets have high life expectancies. Could we lengthen the lives of people in Rwanda by shipping them TV sets? Justify your answer.

LURKING VARIABLE: MONEY



**Problem # 2:**

Do artificial sweeteners cause weight gain? People who use artificial sweeteners in place of sugar tend to be heavier than people who use sugar. Does this mean that artificial sweeteners cause weight gain? Give a more plausible explanation for this association.

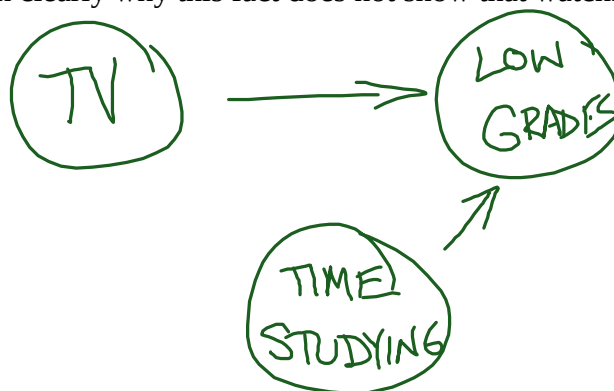
CONFOUNDING: DIET



**Problem # 3:**

Are grades and TV watching linked? Children who watch many hours of television get lower grades in school on the average than those who watch less TV. Explain clearly why this fact does not show that watching TV causes poor grades.

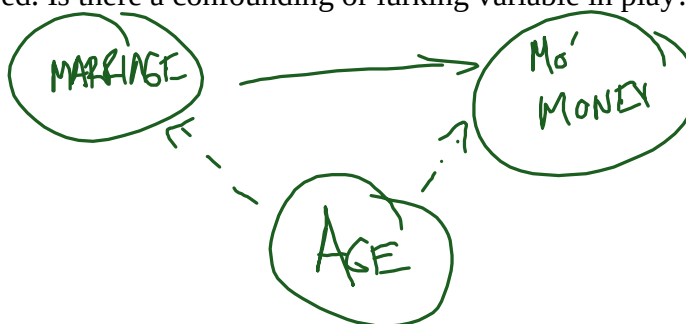
CONFOUNDING: TIME STUDYING



**Problem # 4:**

To earn more, get married? Data show that men who are married, and also divorced or widowed men, earn quite a bit more than men who have never been married. Is there a confounding or lurking variable in play? Explain.

LURKING: AGE

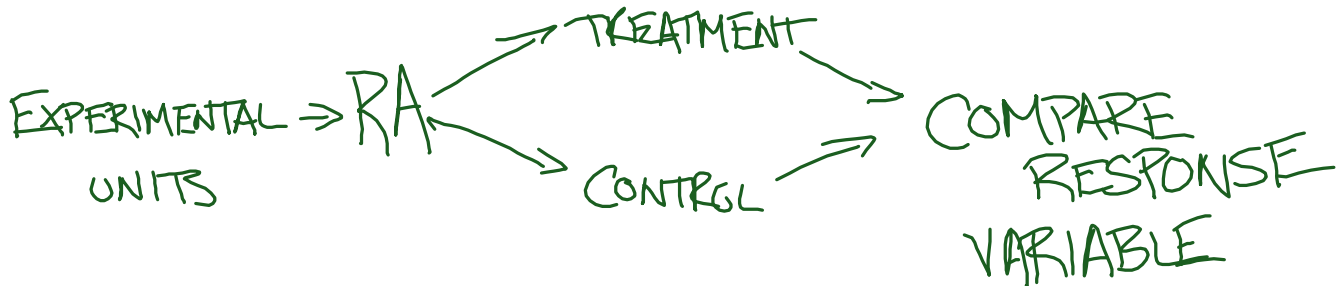


## Experimental Design

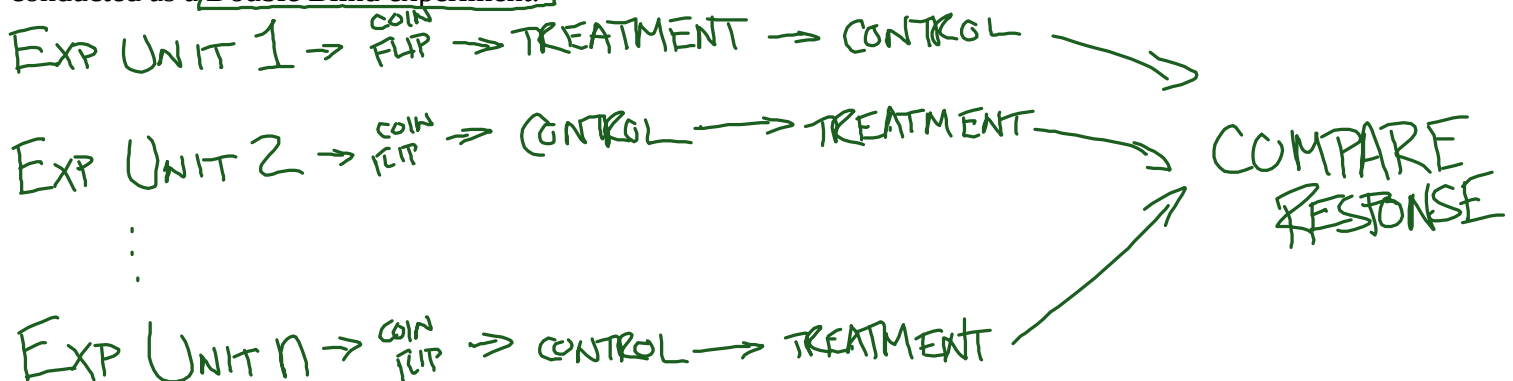
Recap: The design of an experiment is crucial. Experiments are designed with the purpose of isolating the effect of the treatment on the response variable and removing any confounding effects.

One way that we have seen already of removing the effect of any confounding variables is to randomly assign subjects to the treatment or control group. This way any possible bias in the population should be evenly spread among the treatment and control groups. Sometimes instead of relying on randomization to make the groups as even as possible we actually force the groups to be similar.

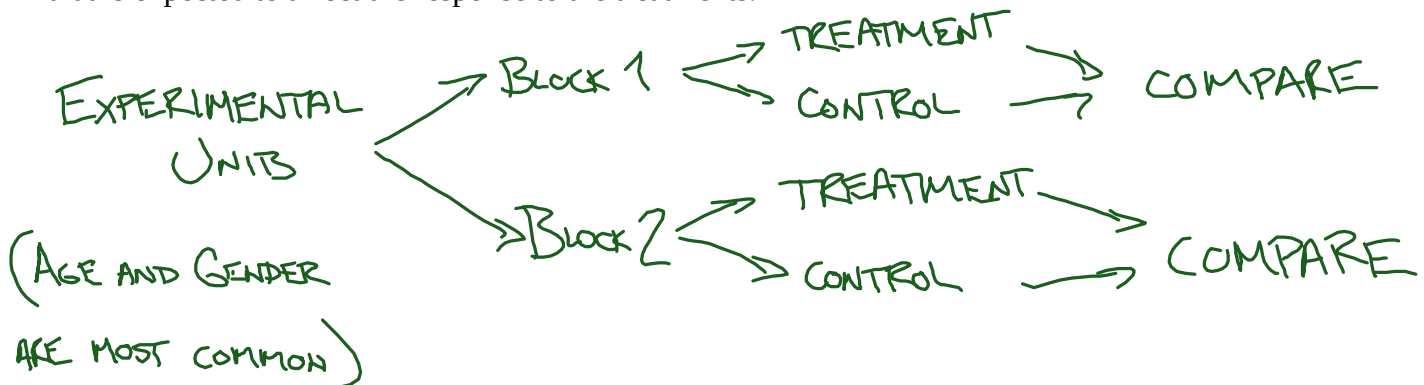
**Completely Randomized Design:** the experimental units are assigned to the treatments completely by chance.



**Matched Pair** designs: These are experimental designs in which either the same individual or two matched individuals are assigned to receive the treatment and the control. In the case where an individual receives both the treatment and the control, the order in which this happens should be random. And the experiment should be conducted as a Double Blind experiment.



**Block Designs:** the random assignment of experimental units to treatments is carried out separately within each block. A block is a group of experimental units that are known before the experiment to be similar in some way that is expected to affect the response to the treatments.



\*\*\*Control what you can, block on what you can't control, and randomize to create compatible groups.

## Scope of Inference

**Inference:** Drawing conclusions beyond the data at hand.

Let's take a look at two different studies:

EX 1) The U.S Census Bureau carries out a monthly Current Population Survey of about 60,000 households. Their goal is to use data from these randomly selected households to estimate the percent of unemployed individuals in the population.

EX 2) Scientists performed an experiment that randomly assigned 21 volunteer subjects to one of two treatments: sleep deprivation for one night or unrestricted sleep. The experimenters hoped to show that sleep deprivation causes a decrease in performance two days later.

- Random selection of individuals SAMPLING ALLOWS INFERENCE FOR THE POPULATION
- Random assignment in an experiment ALLOWS INFERENCE ABOUT CAUSE AND EFFECT

*For the US Census Bureau, individuals were randomly chosen to participate in the survey. The Bureau would be safe in making an inference about the population.*

*In the sleep deprivation experiment, subjects were randomly assigned to their treatments. If there is a large difference in the results, then we can assume it is not due to chance variation between the groups alone, and must be due to sleep deprivation.*

Well-designed experiments randomly assign individuals to treatment groups, but most don't select experimental units at random from the larger population, so their findings are limited to just cause and effect.

| Were individuals randomly assigned to groups? |     |                                                                                            |                                                                                           |
|-----------------------------------------------|-----|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Were individuals randomly selected?           |     | Yes                                                                                        | No                                                                                        |
|                                               | Yes | Inference about the population: <b>YES</b><br>Inference about cause and effect: <b>YES</b> | Inference about the population: <b>YES</b><br>Inference about cause and effect: <b>NO</b> |
|                                               | No  | Inference about the population: <b>NO</b><br>Inference about cause and effect: <b>YES</b>  | Inference about the population: <b>NO</b><br>Inference about cause and effect: <b>NO</b>  |

Both random sampling and random assignment introduce chance variation into a statistical study. When performing inference, statisticians use the laws of probability to describe this chance variation. You'll learn how this works later :)

# **The Challenges of Causation**

In some cases, it is not practical or ethical to do an experiment to establish a cause and effect relationship. Consider the following examples:

- Does texting while driving increase the risk of having an accident?
- Does going to church regularly help people live longer?
- Does smoking cause lung cancer?

The best data we have about these and many other cause and effect questions come from observational studies.

There are laws now that must be followed when dealing with human subjects:

- Reviewed by an institutional review board
- The board's purpose is "to protect the rights and welfare of human subjects (including patients) recruited to participate in research activities".
- Informed consent
- Subjects must be aware of the harm and danger a study could inflict and must given written permission to participate.
- Confidentiality
- Protect individuals' privacy by keeping their identity separate from their results.

## Worst of the worst:

- The Stanford Prison Experiment
- Monkey Drug Trials
- The Wall of Despair
- The Milgram Experiment
- Little Albert